Zeyad Mohamed Salem

QA Automation Engineer · Test Automation, Database & Security Testing, CI
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Available remotely for part-time and contract QA automation work, up to 20 hours per week.

SUMMARY

QA automation engineer and Computer Science student at E-JUST. I write tests that catch real bugs, from Postgres row-level security policies up to the user interface, and run them in CI so broken builds never deploy. I have written more than 2,900 automated tests across two production projects, including red-team suites against privilege escalation. I also build web apps in React, TypeScript and Supabase, so I can read the code I test and find root causes. Co-author of an IEEE NILES 2026 paper.

EDUCATION

B.Sc. Computer Science, Egypt-Japan University of Science and Technology (E-JUST) Expected December 2027

PUBLICATION

A. A. Alshames, I. Abdelmotteleb, M. Sameh, **Z. M. Salem**, S. Farrag, G. Morcos, S. A. Alashry, A. B. Zaky, "AraBIRD: A Modern Standard Arabic Adaptation of BIRD for Text-to-SQL over English Schemas," *2026 IEEE 8th Novel Intelligent and Leading Emerging Sciences Conference (NILES)*. Accepted.

PROJECTS

NC Spark: Role-Based Compliance Training Platform · **github.com/Zeyadmosalem/nc-spark** 2026

Vitest · red-team security tests · GitHub Actions · React 19 · Supabase (Postgres 17, Edge Functions) · Cloudflare Workers

- Wrote 1,828 tests at 91% coverage across 39 migrations and 14 Edge Functions, all running in CI on every push.
- Built red-team security suites that try to escalate privileges and read the quiz answer key, failing the build if either succeeds; escalation is blocked by column grants, row-level security and triggers.
- Verified server-side quiz grading against an answer key no browser role can read, and tested a Cloudflare Worker access gate that checks sessions, throttles sign-ins and sets security headers (CSP, HSTS).
- Built the platform under test: four portals (trainee, trainer, supervisor, admin) and seven authorable activity types.

Samboza: Bilingual Household Finance App (PWA) · **github.com/Zeyadmosalem/Samboza** 2026 · In production

pgTAP · Vitest · GitHub Actions · React 19 · TypeScript · Supabase (Postgres) · Cloudflare Workers

- Proved authorization across four user roles with 532 pgTAP row-level security policy tests, plus 578 front-end tests running in CI on every push.
- Built a CI/CD pipeline across 50 migrations that blocks deploys until the database schema is current.
- Ran the full cycle: interviewed users for requirements, designed the data model, reviewed security, wrote test plans, and deployed.
- Arabic/English app with a full right-to-left layout, used daily by a five-person household since September 2026; double-entry ledger with an audit trail, offline writes through an idempotent outbox.

Beyond Seen Mistakes: Research Pipeline (test & CI contributor) 2026

Python · automated testing · GitHub Actions

- Wrote the test suite and CI for a published research pipeline: data-loader, split-protocol, training edge-case and reproduction tests.
- Fixed unreliable reproduction checks by comparing results numerically instead of bit-for-bit, so they pass consistently across machines.

TRAINING

Mobile App Development Intern, DEPI (Digital Egypt Pioneers Initiative) · Mobile track, final project: a weather app 2024

iOS Development Intern, ITI (Information Technology Institute) · Native iOS development with Swift 2024

SKILLS

Testing: Vitest, pgTAP, Playwright, Testing Library, pytest, red-team security tests, coverage gates, test planning, bug reporting
CI/CD & Tools: GitHub Actions, Git, database migrations, AI-assisted development (Claude Code)
Languages: TypeScript, JavaScript, SQL, Python, C++, Dart, HTML/CSS
Apps & Platforms: React, Vite, Supabase (Postgres, row-level security, Edge Functions), Cloudflare Workers, TanStack Query, PWAs
Spoken: Arabic (native), English (professional working proficiency)